

Cryptographic Tidbits

Lesson 2

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Cryptographic Tidbits

Lesson 2

- Codes vs ciphers

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- Codes vs ciphers
- Transposition vs Substitution

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- Codes vs ciphers
- Transposition vs Substitution
- Monoalphabet vs Polyalphabetic

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- Codes vs ciphers
- Transposition vs Substitution
- Monoalphabet vs Polyalphabetic
- Cryptanalysis

Codes and ciphers are NOT the same

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Codes and ciphers are NOT the same

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- Code
 - Word/phrase/concept replacement

Codes and ciphers are NOT the same

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- **Code**
 - **Word/phrase/concept replacement**
 - **One if by land, two if by sea.**

Codes and ciphers are NOT the same

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- **Code**
 - **Word/phrase/concept replacement**
 - **One if by land, two if by sea.**
- **Cipher**
 - **Character/symbol replacement**

Historical ciphers reveal basic concepts

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- Transposition ciphers
 - The scytale

Historical ciphers reveal basic concepts

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- Transposition ciphers
 - The scytale
- Substitution ciphers
 - Caesar
 - Mary Queen of Scots
 - Vigenere

The Scytale

The first cipher used transposition

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The Caesar Ciphers

The first substitution ciphers

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The Caesar Ciphers

The first substitution ciphers

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- Caesar Cipher
 - Latin with Greek

The Caesar Ciphers

The first substitution ciphers

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- Caesar Cipher
 - Latin with Greek
- Caesar Shift Cipher
 - Shift alphabet by 3



Mary Queen of Scots

Cryptanalysis comes to Europe

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The Vignere Cipher

“Unbreakable” now broken as homework

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Lesson 2 in Review

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- **Coming Up**
 - **Machine-based systems**

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 - **Mathematically provable security**

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 - **Computational security**
 - **Mathematically provable security**
 - **Breaking the unbreakable**